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KELAS : SIB-2G
MATERI : Percabangan, Perulangan dan Operasi Git Lanjutan pada Dart

Studi Kasus: Aplikasi Kalkulator BMI

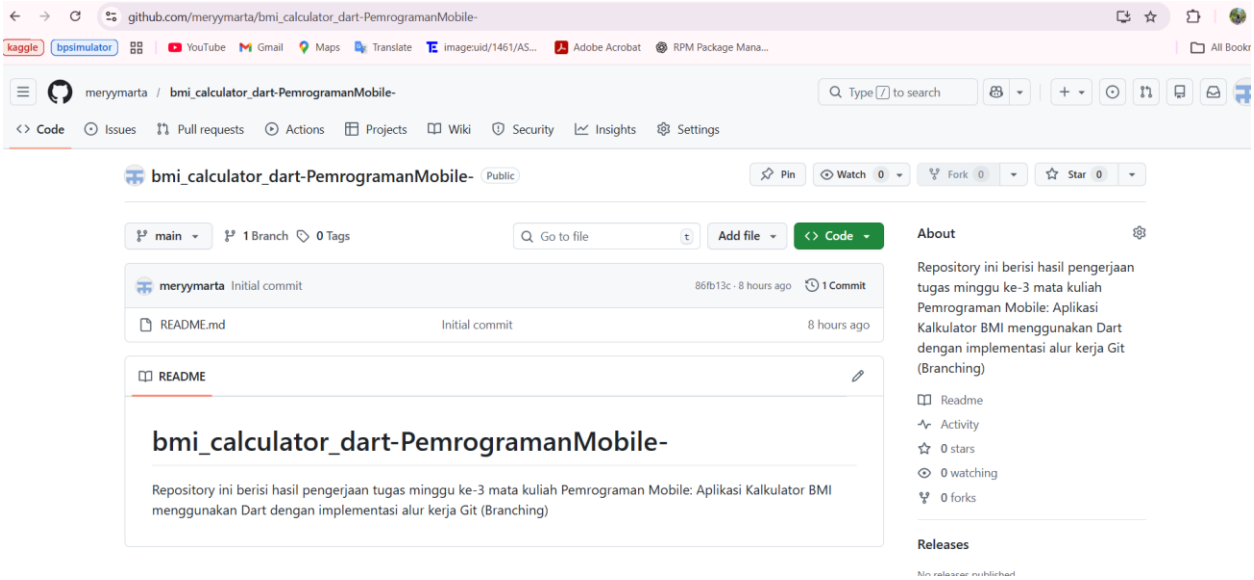
Kita akan mengembangkan aplikasi kalkulator BMI sederhana menggunakan Dart dengan menerapkan percabangan dan perulangan.

Fitur Aplikasi:

- Input tinggi dan berat badan
- Kalkulasi BMI
- Kategorisasi hasil (kurus, normal, gemuk)
- Riwayat perhitungan (menggunakan perulangan)

Alur Kerja Git

1. Buat Repository baru di Github



2. Clone repository ke local
3. Buat branch untuk setiap fitur
   - Branch input-validation

```
3. Tinggi: 180.0 cm, Berat: 85.0 kg | BMI: 26.23 -> Gemuk
PS C:\Users\TOSHIBA\bmi_calculator_dart-PemrogramanMobile-\bmi_calculator_dart> git checkout input-validation
Switched to branch 'input-validation'
Your branch is up to date with 'origin/input-validation'
```

   - Bracnh bmi-calculation

```
PS C:\Users\TOSHIBA\bmi_calculator_dart-PemrogramanMobile-> git checkout -b bmi-calculation
Switched to a new branch 'bmi-calculation'
```

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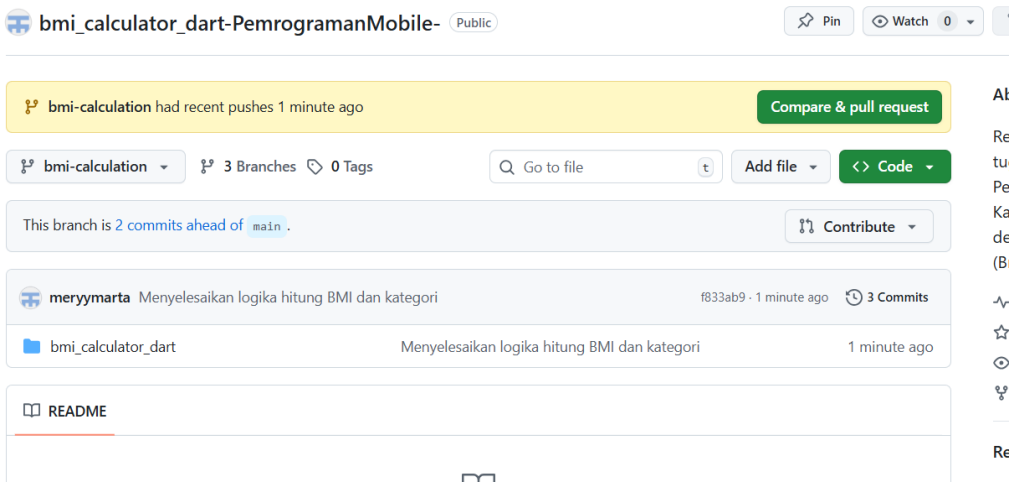
```dart
void hitungBMI(double tinggiCm, double beratKg, List riwayat) {
  // Konversi tinggi dari cm ke meter
  double tinggiM = tinggiCm / 100;

  // Hitung BMI
  double bmi = beratKg / (tinggiM * tinggiM);
  //menentukan kategori menggunakan percabangan
  String kategori;
  if (bmi < 18.5) {
    kategori = "Kurus";
  } else if (bmi < 25) {
    kategori = "Normal";
  } else if (bmi < 30) {
    kategori = "Gemuk";
  } else {
    kategori = "Obesitas";
  }

  riwayat.add({
    "tinggi": tinggiCm,
    "berat": beratKg,
    "bmi": bmi,
    "kategori": kategori
  });

  print("Data masuk -> Tinggi: $tinggiCm cm, Berat: $beratKg kg");

}
```

```
PS C:\Users\TOSHIBA\bmi_calculator_dart-PemrogramanMobile-> git add .
warning: in the working copy of 'bmi_calculator_dart/bin/main.dart', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\TOSHIBA\bmi_calculator_dart-PemrogramanMobile-> git commit -m "Menyelesaikan logika hitung BMI dan kategori"
[bmi-calculation f833ab9] Menyelesaikan logika hitung BMI dan kategori
 1 file changed, 25 insertions(+)
PS C:\Users\TOSHIBA\bmi_calculator_dart-PemrogramanMobile-> git push -u origin bmi-calculation
```



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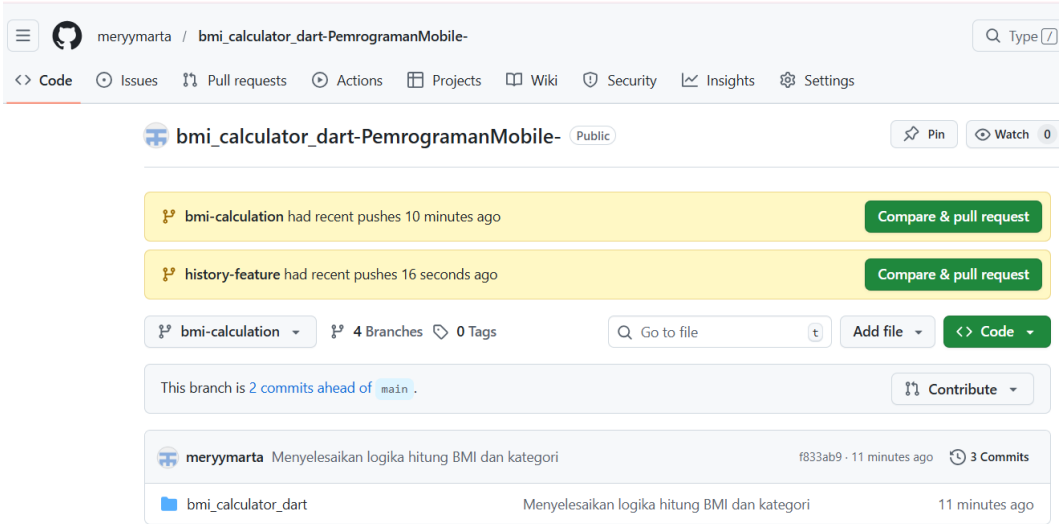
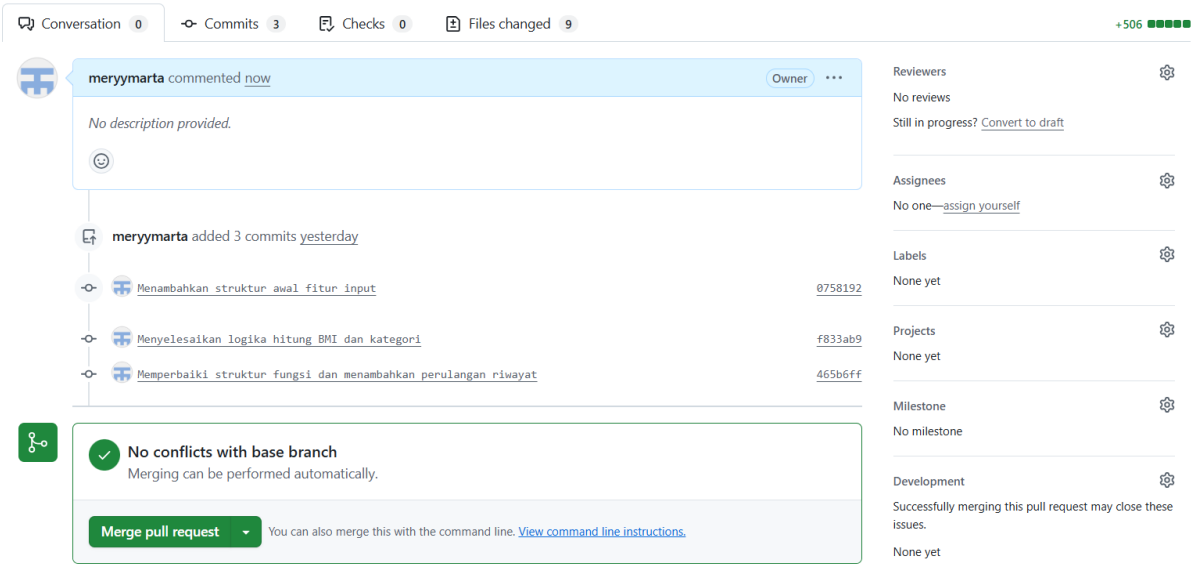
- Branch history-feature

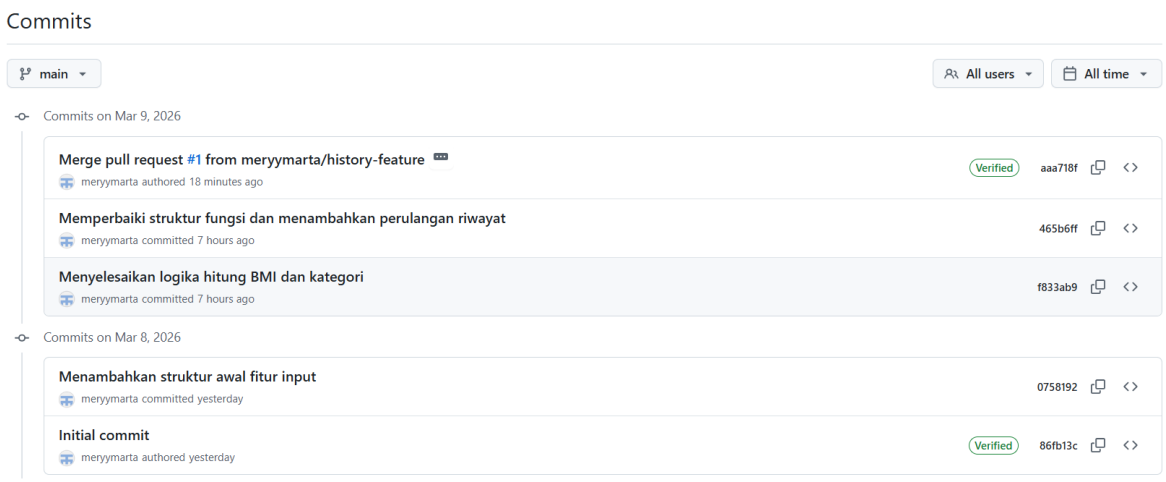
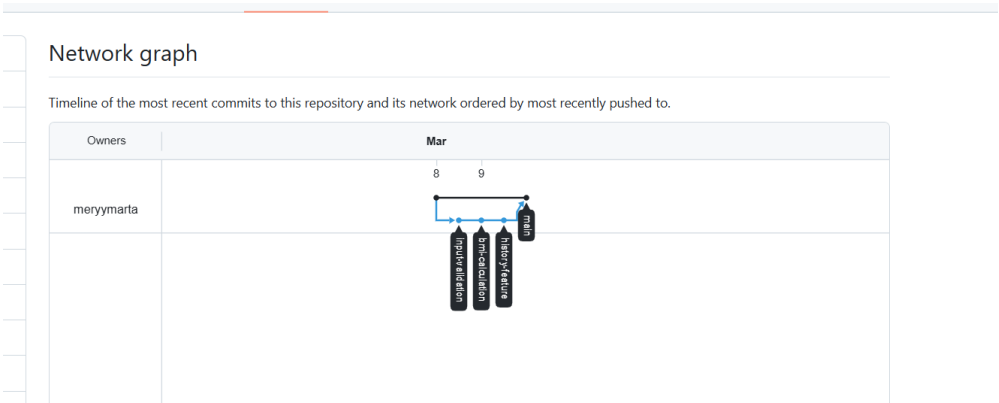
```
PS C:\Users\TOSHIBA\bmi_calculator_dart-PemrogramanMobile-> git checkout -b history-feature
Switched to a new branch 'history-feature'
```

```
PS C:\Users\TOSHIBA\bmi_calculator_dart-PemrogramanMobile-> git add .
warning: in the working copy of 'bmi_calculator_dart/bin/main.dart', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\TOSHIBA\bmi_calculator_dart-PemrogramanMobile-> git commit -m "Memperbaiki struktur fungsi dan menambahkan perulangan riwayat"
[history-feature 465b6ff] Memperbaiki struktur fungsi dan menambahkan perulangan riwayat
 1 file changed, 20 insertions(+), 5 deletions(-)
PS C:\Users\TOSHIBA\bmi_calculator_dart-PemrogramanMobile-> git push -u origin history-feature
```



4. Merge branch ketika fitur selesai



5. Push ke repository remote

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